Arshie Fatima
Farmington Hills, Michigan, USA

Mobile:248-956-0644

Email: Arshie.fatima@gmail.com
https://github.com/arshiefatima
www.linkedin.com/in/arshie-fatima-1707361ab

Professional Summary

AI Engineer | LLM Systems | RAG Pipelines | Multi-Agent Systems | LangChain | LangGraph | Llama | API Integration | LLM Evaluation | Production AI

AI Engineer specializing in LLM-based applications, RAG pipelines, and agentic automation systems. Experienced in designing scalable AI workflows using LangChain, LangGraph, vector databases, and API integrations. Strong background in test automation and system reliability, now focused on building intelligent AI-driven automation solutions, multi-agent systems, and production-grade LLM applications. Skilled in Python, NLP, and end-to-end AI system design.

Transitioned from QA Automation Engineer to AI Engineer focusing on LLMs, RAG systems, and agentic workflows.

Technical Skills

- **AI/LLM:** LLMs, RAG, LangChain, LangGraph, Prompt Engineering, NLP, LLM Evaluation
- **Programming Languages:** Python, Core Java, C, C++
- **Web Technologies:** JavaScript, HTML, JSON, CSS.
- **Database:** MySQL, MS SQL Server, SQL Server Management Studio, Oracle 10g.
- **Test Management Tools:** Jira, HP ALM, Test Rail etc.,
- **Browsers:** Internet Explorer, Mozilla Firefox, Google Chrome, Safari
- **Programming IDE's:** Eclipse, Visual Studio, IntelliJ IDE
- **OS:** Windows, Mac iOS, Linux
- **Other:** REST APIs, CI/CD, GitHub, Agile/Scrum

Work Authorization: Authorized to work in USA possessed Employment Authorization Document

Work Experience

Walmart — Mississauga, Canada — March 2024– Till now

AI Engineer - Remote

Responsibilities

- Developed AI-powered automation workflows using LangChain and LangGraph to streamline retail and supply chain operations, reducing manual effort by ~30%
- Built and optimized RAG-based systems using OpenAI embeddings and Pinecone vector database for enterprise knowledge retrieval
- Designed multi-agent AI architectures to automate end-to-end business workflows through coordinated agent execution
- Integrated LLM-based systems with external REST APIs for real-time data processing and automation
- Implemented Playwright-based automation frameworks for web workflow testing and execution
- Used LangSmith for monitoring and debugging LLM outputs, improving response reliability and reducing failure rate
- Built reusable AI components using LCEL (LangChain Expression Language) for scalable orchestration
- Integrated AI agents into CI/CD pipelines for automated unit test generation, error resolution, and environment setup, utilizing tools like GitHub Copilot and Playwright for E2E automation
- Tuned embedding chunking strategies (size, overlap, metadata tagging) to improve relevance and reduce noise in retrieved results by ~25–30% in enterprise knowledge search workflows
- Reviewed and debugged AI-generated and production code across Python and Java, identifying edge cases, performance issues, and logic flaws to ensure production-quality output
- Reduced manual operational effort by ~30% through LLM-powered automation workflows

- Improved retrieval accuracy by ~25–30% via embedding and chunking optimization

Zighra Ontario, Canada

AI Engineer-Intern Nov2023-Jan2024

- Built and optimized LLM-powered applications using Python, focusing on prompt engineering to improve response accuracy, relevance, and consistency across different user queries.
- Designed and implemented RAG (Retrieval-Augmented Generation) pipelines to enhance LLM outputs by integrating external knowledge sources for context-aware responses.
- Worked with vector databases like Pinecone to store and retrieve embeddings efficiently for semantic search and intelligent question-answering systems.
- Experimented with embedding models and chunking strategies to improve retrieval quality and reduce hallucinations in LLM responses

Intact Financial Corp Toronto, Canada

QA Analyst(Automation Focused) June 2021– Oct2023

Responsibilities

- Developed and maintained automation scripts using Selenium and Java, improving test coverage and reducing regression time by ~40%
- Performed functional, regression, API, and UI testing across web applications
- Automated API testing using Postman and integrated with CI/CD pipelines
- Executed SQL queries for backend data validation and ensured data accuracy
- Worked in Agile teams, collaborating with developers and business analysts for defect resolution
- Contributed to test planning, execution, and reporting in fast-paced release cycles

Quality Analyst- Intern June 2020– May 2021

- Created and executed test cases and supported regression testing
- Participated in Agile ceremonies and defect tracking using Jira
- Assisted in manual and automation testing across multiple applications
- Helped in writing, reviewing, and executing manual test cases
- Participated in requirement discussions and Agile/Scrum meetings with the team.

Projects

Multi-Agent AI Analyst System

- Built a multi-agent AI system using LangGraph and LangChain to automate complex task workflows through agent collaboration
- Designed modular agents for task planning, execution, and response generation using LLM-based reasoning
- Integrated external APIs and tools to enable real-time data processing and automated decision-making
- Developed a scalable orchestration framework with state management for reliable multi-agent communication

Tech Stack: Python, LangChain, LangGraph, LLMs, OpenAI APIs, Llama.

GitHub: https://github.com/arshiefatima/local-multi-agent-ai-analyst

AI Autonomous Research Agent

• Developed a multi-agent research system for autonomous query planning, web/tool-based research, and report synthesis
• Implemented LLM-driven agent workflows for task decomposition and structured insight generation
• Built tool-calling and API integration for real-time information retrieval and reasoning

Tech Stack: Python, LangChain, LangGraph, OpenAI APIs, Tool Calling, LLMs

GitHub: https://github.com/arshiefatima/ai-autonomous-research-agent

Intelligent Automation Agent (LLM + API + RAG System)

- Built an AI-powered automation agent that performs task understanding, planning, and execution using LLM-based reasoning
- Integrated external APIs and tools to enable dynamic decision-making and real-time data-driven responses
- Designed a modular pipeline for automation workflows including summarization, email generation, and task classification
- Implemented intent detection and orchestration logic to route user queries into appropriate AI-driven actions

Tech Stack: Python, LangChain, LLMs, OpenAI APIs, REST APIs, NLP

GitHub: https://github.com/arshiefatima/intelligent-automation-agent

Achievements

Received Graduate Achievement Award at St. Clair College for my outstanding academic performance.

Education

Post Graduate in Data Analytics for Business *St. Clair College, Windsor, ON, Canada* **May 2021**
GPA: 3.53
Bachelors in Information Technology *Stanley College of Engineering* **May 2019**
GPA: 3.53